Unit: Unit 0: Foundations of Algebra
Topic: Rational Numbers Decimals

Name: ____________

Date: ________

Quiz (Carolina Reaper)

Q1. A shipping company has $3\frac{1}{4}$ hours to deliver a package. What is this time in decimal form?
(A) 3.1
(B) 3.25
(C) 3.5
(D) 3.75
(E) 4.0

Q2. What is the value of x in the equation $2 \cdot x = 1\frac{3}{4}$?
(A) 0.75
(B) 0.875
(C) 1.0
(D) 1.25
(E) 1.5

Q3. A traffic flow study found that the average commute time was $45\frac{1}{2}$ minutes. What percent of an hour is this?
(A) 72.5
(B) 75
(C) 76.25
(D) 80
(E) 82.5

Q4. A delivery truck travels $2\frac{3}{8}$ miles in $\frac{1}{4}$ hour. What is its average speed in miles per hour?
(A) 8
(B) 9.5
(C) 10
(D) 12
(E) 16

Q5. What is the decimal equivalent of $\frac{5}{8}$?
(A) 0.5
(B) 0.6
(C) 0.625
(D) 0.75
(E) 0.8

Q6. A logistics company has a $3:2$ ratio of trucks to drivers. If there are $4\frac{1}{2}$ drivers, how many trucks are there?
(A) 5
(B) 6
(C) 6.75
(D) 7
(E) 8

Q7. A package delivery service guarantees $98\frac{3}{4}$
(A) 0.983
(B) 0.984
(C) 0.985
(D) 0.9875
(E) 0.988

Q8. What is the value of x in the equation $\frac{2}{3}x = 1\frac{1}{8}$?
(A) 1.5
(B) 1.75
(C) 2.0
(D) 2.25
(E) 2.5

Open Ended Questions (FRQ)

Q9. A traffic engineer wants to represent the average commute times on a number line, with values ranging from 0 to 2 hours. If the average commute time is $1\frac{1}{4}$ hours, where would it be placed on the number line? Show your work and explain your answer.

Q10. A shipping company has a $2:5$ ratio of overnight to standard deliveries. If $4\frac{3}{8}$ standard deliveries were made, how many overnight deliveries were made? Show your work and explain your answer.

Chef's Tips

- Chef's Tips: Ensure students show their work for FRQs
- Chef's Tips: Encourage students to use number lines for questions involving rational numbers
- Chef's Tips: Emphasize the importance of converting between fractions, decimals, and percents in real-world applications